4	**(a)**	The current carrying wire experiences a magnetic force in the vertical plane as given by Fleming’s Left Hand Rule. Since the current is alternating and changes direction, the direction of force alternates and causes the wire to vibrate. The vibration will cause progressive waves travelling towards P & M and the reflected waves will superpose. Resonance occurs and a stationary wave will be formed when the frequency of the periodic magnetic force (due to the sinusoidal current) matches the natural frequency of the stationary waves in the string.	
	(b)	$0.6 = \frac{\lambda}{2}$ $\lambda = 1.20$ m	
	(c)	v = f $\lambda = (10)1.20$ = 12.0 m s^{-1}	
	(d)	Correctly drawn stationary wave with 3 antinodes and 4 nodes, with 2 of the nodes at M and P. (Three loops with nodes at the ends	

		with equal distance between adjacent nodes) Reason: Same tension, speed is the same Determine new wavelength when freq is increased to 30 Hz. $\lambda = \frac{v}{f} = \frac{12}{30} = 0.4\ m$ Number of wavelengths formed = 0.6/0.4 = 1.5 There will be 3 loops formed.	

5	**(a)**	**(i)**	$F = Bqv$	